Computer Science Concepts

Memory

Types of Memory: Primary, Secondary, Cache

- Detailed Answer: Primary Memory (RAM) computer ki temporary storage hai jahan data tab tak rakha jata hai jab tak computer on hota hai. Yeh fast hota hai lekin iska data power band hone par chala jata hai. Secondary Memory (HDD/SSD) long-term storage hai jahan data permanently rakha jata hai, jaise ki aapke documents aur files. Cache Memory processor ke pass hoti hai aur frequently accessed data ko store karti hai, jisse speed badh jati hai.
- Short Answer: Primary memory RAM hoti hai, secondary memory HDD/SSD hoti hai, aur cache memory fast access ke liye hoti hai.
- Example: Jab aap ek document open karte hain, toh yeh pehle RAM me load hota hai, phir agar aap save karte hain toh yeh HDD me store hota hai.

Processors

What is a CPU and how does it work?

- Detailed Answer: CPU (Central Processing Unit) computer ka brain hota hai. Yeh instructions ko process karta hai, calculations karta hai aur computer ke tasks ko manage karta hai. CPU ke andar alag-alag cores hote hain jo parallel processing me madad karte hain.
- Short Answer: CPU computer ka brain hota hai jo tasks ko process karta hai.
- Example: Jab aap kisi software ko run karte hain, CPU us software ke instructions ko process karta hai.

What are Intel and AMD processors?

- Detailed Answer: Intel aur AMD do main companies hain jo processors banati hain. Intel

processors stable hote hain aur gaming aur professional tasks ke liye popular hain, jabki AMD processors zyada cores aur better multi-threading performance dete hain.

- Short Answer: Intel aur AMD processors computer ke liye important hote hain, dono ke apne features hain.

- Example: Aapke laptop me Intel processor ho sakta hai jo smooth performance deta hai ya AMD processor jo gaming ke liye better hota hai.

CPU Architecture

What are instruction sets in processors?

- Detailed Answer: Instruction sets CPU ke liye ek command list hoti hai jo CPU ko batati hai ki kaise tasks perform karne hain. Yeh commands CPU ko samajh aati hain aur unhe execute karti hain.

- Short Answer: Instruction sets CPU ke liye commands ka set hota hai.

- Example: Jab aap ek game khelte hain, toh CPU ko instructions chahiye hoti hain jaise "move forward" ya "jump."

Storage

What is the difference between HDD and SSD?

- Detailed Answer: HDD (Hard Disk Drive) mechanical parts ke sath hota hai, jo data read/write karne ke liye slow hota hai. SSD (Solid State Drive) flash memory use karta hai, jo zyada fast hota hai aur shock-resistant hota hai.

- Short Answer: HDD slow hota hai aur SSD fast hota hai.

- Example: Agar aapka computer HDD use karta hai, toh boot hone me zyada time lagega, jabki SSD use karne par jaldi boot hoga.

Computer Science Concepts

Networking

What is Ethernet and how does it work?

- Detailed Answer: Ethernet wired networking technology hai jo computers ko local area network (LAN) me connect karta hai. Yeh data ko packets me transmit karta hai aur isme cables ka use hota hai.

- Short Answer: Ethernet wired connection hai jo network devices ko connect karta hai.

- Example: Aapka home Wi-Fi router agar Ethernet cable se connected hai, toh yeh stable internet provide karega.

Internet

What is an IP address and why is it important?

- Detailed Answer: IP address har device ko internet par unique identification deta hai. Yeh device ko locate karne me madad karta hai taaki data sahi jagah par ja sake.

- Short Answer: IP address device ka unique identifier hota hai internet par.

- Example: Jab aap kisi website ko access karte hain, aapke device ko IP address ki madad se website tak pahunchaya jata hai.

Hardware Components

What is a motherboard and why is it essential?

- Detailed Answer: Motherboard computer ka main circuit board hai jahan sabhi components connect hote hain, jaise CPU, RAM, aur storage devices. Yeh sabhi parts ko communicate karne me madad karta hai.

- Short Answer: Motherboard sabhi components ko connect karta hai aur essential hai.

- Example: Jab aap apne computer me hardware upgrade karte hain, motherboard hi sabhi new components ko manage karta hai.

Peripherals and I/O Devices

What are input and output devices?

- Detailed Answer: Input devices woh hote hain jo user se data lete hain, jaise keyboard aur mouse. Output devices woh hain jo processed data ko user tak pahunchate hain, jaise monitor aur printer.

- Short Answer: Input devices data lete hain aur output devices data dikhate hain.

- Example: Aap keyboard se data type karte hain (input) aur monitor par dekhte hain (output).

Mobile Technology

What is Snapdragon used for in mobile devices?

- Detailed Answer: Snapdragon processors mobile devices me use hote hain aur yeh performance aur battery efficiency ke liye famous hain. Inme built-in GPUs bhi hote hain jo gaming aur multimedia performance improve karte hain.

- Short Answer: Snapdragon mobile processors hote hain jo performance ke liye ache hain.

- Example: Aapka smartphone agar Snapdragon processor use karta hai, toh yeh smooth gaming experience provide karega.

Operating Systems

What is the difference between Windows and Linux?

- Detailed Answer: Windows ek proprietary operating system hai, jise Microsoft ne develop kiya hai, jabki Linux open-source hai, jo free me use kiya ja sakta hai. Linux zyada customizable hota hai aur developers ke beech popular hai.
- Short Answer: Windows proprietary hai, jabki Linux open-source hai.
- Example: Agar aap ek gamer hain, toh aap Windows use karte hain, lekin agar aap programmer hain, toh Linux aapke liye better choice ho sakta hai.

Software and Applications

What is the difference between system software and application software?

- Detailed Answer: System software computer hardware ko manage karta hai, jaise operating systems. Application software specific tasks perform karta hai, jaise word processors aur games.
- Short Answer: System software hardware ko manage karta hai, application software tasks perform karta hai.
- Example: Windows (system software) aapke computer ko manage karta hai, aur Microsoft Word (application software) document banane ke liye use hota hai.

Networking Devices

What is a router and how does it work?

- Detailed Answer: Router ek networking device hai jo data packets ko network me route karta hai. Yeh devices ko connect karta hai aur internet access provide karta hai.
- Short Answer: Router data ko network me route karta hai aur internet access provide karta hai.
- Example: Aapka home router aapke internet connection ko aapke devices tak pahunchata hai.

Computer Science Concepts

Cloud Computing

What is cloud computing?

- Detailed Answer: Cloud computing internet ke zariye data storage aur processing ko refer karta hai. Users data ko remotely store kar sakte hain aur kabhi bhi access kar sakte hain, bina physical storage device ke.
- Short Answer: Cloud computing remote data storage aur processing hai.
- Example: Google Drive ya Dropbox me aap apne files store karte hain, jo aap kahin bhi access kar sakte hain.

Data Transmission

How does Bluetooth work for data transmission?

- Detailed Answer: Bluetooth ek wireless technology hai jo short-range data transmission ke liye use hoti hai. Yeh devices ko connect karta hai aur unke beech data share karta hai bina cables ke.
- Short Answer: Bluetooth wireless data transmission ke liye hota hai.
- Example: Aap apne smartphone se wireless headphones ko connect karte hain Bluetooth ka use karke.

Emerging Technologies

Emerging Technologies

What is artificial intelligence (AI)?

- Detailed Answer: Artificial intelligence (AI) computer systems ko aise tasks perform karne ke liye

enable karta hai jo human intelligence ki zarurat rakhte hain, jaise decision making, problem-solving aur learning.

- Short Answer: AI computer systems ko human-like tasks perform karne ke liye enable karta hai.
- Example: Aapke smartphone me voice assistants jaise Siri ya Google Assistant AI ka istemal karte hain.

What is the Internet of Things (IoT) and how is it transforming devices?

- Detailed Answer: IoT devices internet se connected hote hain aur ek dusre se communicate karte hain. Yeh devices smart homes, healthcare aur industries me efficiency badhate hain.
- Short Answer: IoT devices interconnected hote hain aur daily life me efficiency badhate hain.
- Example: Smart thermostats jo aapke ghar ki temperature ko automatically adjust karte hain woh IoT devices hain.

What is blockchain technology?

- Detailed Answer: Blockchain ek distributed ledger technology hai jo transactions ko secure aur transparent tarike se record karta hai. Yeh tampering se bacha raha hai aur trust establish karta hai.
- Short Answer: Blockchain secure aur transparent transaction recording ke liye use hota hai.
- Example: Bitcoin jaise cryptocurrencies blockchain technology ka istemal karte hain taaki transactions ko secure rakha ja sake.

Security

What is encryption and why is it important for data security?

- Detailed Answer: Encryption data ko secure karne ka process hai jisme data ko ek code me convert kiya jata hai taaki sirf authorized log hi use padh sakein. Yeh data ko unauthorized access

se bachata hai.

- Short Answer: Encryption data ko secure karta hai taaki sirf authorized log hi access kar sakein.

- Example: Jab aap online banking karte hain, toh aapka data encryption ke through secure hota hai, jisse hackers use nahi padh sakte.

How do firewalls protect networks?

- Detailed Answer: Firewalls ek security system hai jo network traffic ko monitor aur control karta hai. Yeh unauthorized access ko rokne ke liye rules set karta hai, jisse hackers ya malicious software network me nahi aa sakte.

- Short Answer: Firewalls network ko unauthorized access se protect karte hain.

- Example: Aapka home router me firewall hota hai jo aapke network ko secure karta hai jab aap internet use karte hain.

Energy Efficiency in Tech

What factors affect battery life in devices?

- Detailed Answer: Battery life kai factors par depend karti hai jaise screen brightness, running apps, connectivity (Wi-Fi, Bluetooth), aur device ke hardware. Zyada demanding apps ya high brightness battery ko jaldi khatam kar dete hain.

- Short Answer: Screen brightness, apps, aur connectivity battery life ko affect karte hain.

- Example: Agar aap gaming kar rahe hain toh aapki device ki battery jaldi discharge hogi as compared to jab aap sirf web browsing kar rahe hain.

How do processors contribute to energy efficiency?

- Detailed Answer: Processors energy efficiency ko improve karne ke liye advanced technology use

karte hain, jisse performance increase hoti hai bina zyada energy use kiye. For example, modern processors low power states me ja sakte hain jab idle hote hain, jisse battery life badh jaati hai.

- Short Answer: Processors advanced technology use karke energy efficiency ko improve karte hain.
- Example: Aapka laptop agar Intel ka latest processor use karta hai, toh yeh energy-efficient hai aur aapko zyada battery backup milega.